

Choosing the Facial Emotion Model

AI Actor Coach (ActorPro AI) · Final Year Project · October 2026

1. What we needed

While a student practises a line, the app watches their face through the webcam and checks whether it shows the right emotion. The model must recognise **seven emotions** (anger, disgust, fear, joy, neutral, sadness, surprise), work on **ordinary webcam faces**, and be **fast enough to run in the web browser**.

2. Datasets we tested on

We tested every model on the same four datasets. Each one represents a different situation the app meets: everyday faces, clear acted expressions, low-quality pictures, and actors performing lines on video. A model was only scored on a dataset it had **not** been trained on, so no model could pass from memory.

RAF-DB Real-world faces · our main test

Type of faces	Real, everyday photos of people from the internet: all ages, genders and ethnicities, with natural (not acted) expressions.
Conditions	Uncontrolled: different lighting, head angles, glasses, hair and backgrounds, much like a student's webcam at home.
Pictures	Colour photos; faces aligned and cropped to 100×100 pixels.
Labels	Each photo was labelled by about 40 people, and the emotion they agreed on was kept.
Full dataset	About 30,000 photos; around 15,000 show one of the 7 basic emotions.
What we used	1,800 photos : 300 each of anger, disgust, fear, joy, sadness and surprise.
Why it matters	It is the closest match to how the app is really used, so it decided the choice.

CK+ Posed expressions · closest to acting

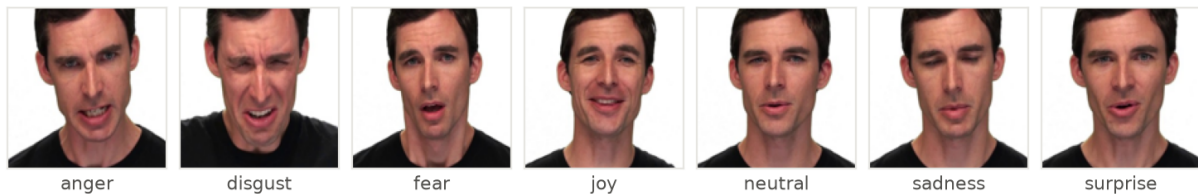
Type of faces	Adult volunteers (aged 18–50) asked to act out each emotion, recorded in a laboratory.
Conditions	Controlled: face looking straight at the camera, even lighting, plain background.
Pictures	Black-and-white face crops, 48×48 pixels, taken at the strongest point of each expression.
Full dataset	593 video sequences from 123 people.
What we used	927 faces showing anger, disgust, fear, joy, sadness and surprise (contempt left out, as the app does not use it).
Why it matters	Expressions are clear and exaggerated, like an actor's, so it shows how well a model reads deliberate acting.

FER-2013 Low-quality internet faces

Type of faces	Faces collected automatically from Google image searches for emotion words: real people, many styles.
Conditions	Mixed and often poor: small, blurry, sometimes partly hidden faces.
Pictures	Black-and-white, only 48×48 pixels.
Labels	Known to be noisy: even people agree with the given label only about 65% of the time.
Full dataset	35,887 faces; the official test part has 7,178.
What we used	1,000 faces picked at random from the test part, covering all 7 emotions.
Why it matters	A hard, low-quality test. It also exposed models that had already seen these test photos during training.

RAVDESS Acted video · used for the final check

Type of faces	24 professional actors (12 female, 12 male) speaking short lines while acting an emotion.
Conditions	Studio recording: good lighting, actor facing the camera, at normal and strong intensity.
Pictures	Colour HD video (1280×720). We took one frame every 0.35 seconds, the same rate as the app.
Full dataset	7,356 recordings with 8 emotions (calm was counted as neutral).
What we used	240 video clips from 4 actors (2 female, 2 male): 2,666 frames covering all 7 emotions.
Why it matters	It is the closest to what the app does: watching someone's face while they perform a line.



Examples from RAVDESS (faces cut from video frames).

3. Candidate models and results

Model	Trained on	Real-world (RAF-DB)	Posed (CK+)	FER-2013	Speed per face
HSEmotion EfficientNet-B2 (chosen)	AffectNet	57.4%	85.8%	56.8%	14 ms
HSEmotion EfficientNet-B2, 8-class	AffectNet	59.7%	82.6%	57.6%	14 ms
mo-thecreator ViT	FER-2013, MMI, AffectNet	52.2%	68.4%	not counted*	49 ms
HardlyHumans ViT	FER-2013, AffectNet	44.5%	77.1%	44.1%	47 ms
trpakov ViT	FER-2013	43.9%	66.7%	71.1%	49 ms
prithivMLmods SigLIP2	not stated	40.4%	48.3%	45.9%	62 ms
dima806 ViT	not stated	38.6%	65.8%	not counted*	48 ms

Scores show the share of faces where the model named the correct emotion (each emotion counted equally). Guessing at random would score about 17%. * These two models had almost certainly seen the FER-2013 test photos during training (they scored around 90%, higher than any published result), so that number does not count.

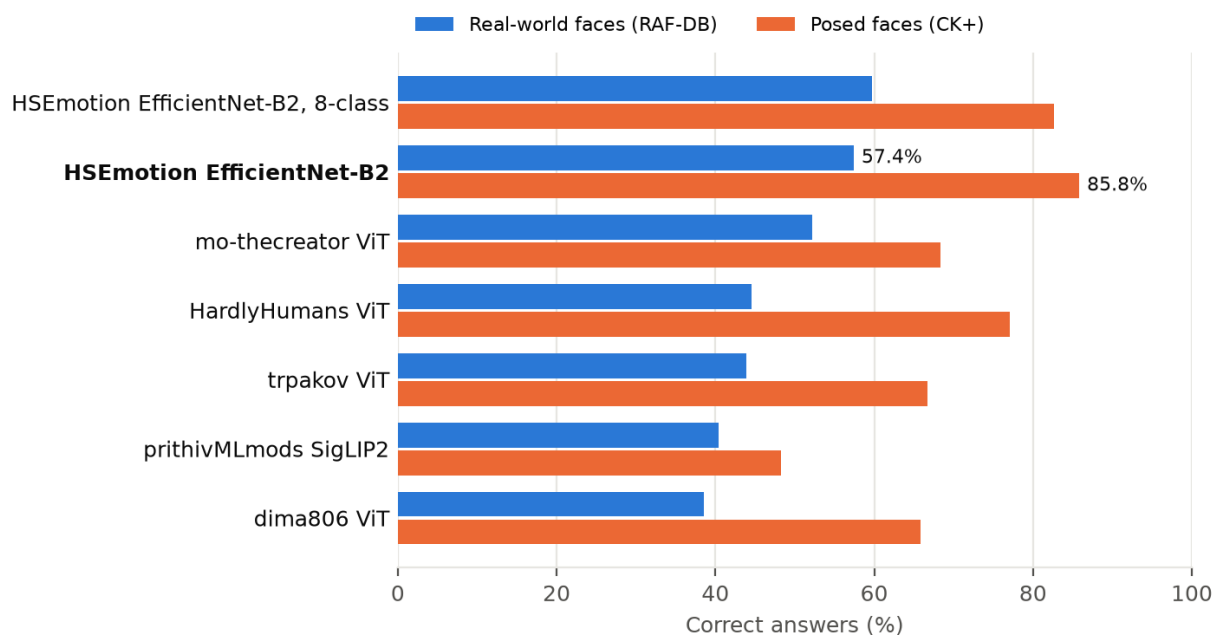


Figure 1. How often each model named the correct emotion. The chosen model is in bold.

4. Why we chose HSEmotion EfficientNet-B2

- **Best on posed faces (85.8%)**, the kind of clear expression an actor makes.
- **Among the best on real-world faces (57.4%)**, about 5 points ahead of the next model type (52.2%). Its 8-class version scored slightly higher here (59.7%), but the two are within each other's margin of error, and it was weaker on posed faces.
- **Matches the app exactly**: it outputs the same seven emotions the app uses, with no extra classes to remove.
- **Fair result**: it was trained on a different dataset (AffectNet), so none of our test faces were seen before.
- **Small and fast**: 8 million parameters and about 14 ms per face, around 3–4 times faster than the larger ViT models. It runs inside the browser, so **the webcam video never leaves the student's computer**.
- **Free to use**: Apache-2.0 licence.

5. Final check in the app

We added the chosen model to the app and tested it on the RAVDESS acted videos (240 clips). It named the correct emotion **71% of the time**. In a live browser test it correctly read an angry, a happy and a neutral performance. Its weakest emotion is subtle anger in everyday photos.

Credits: HSEmotion by A. V. Savchenko (Apache-2.0). Datasets: RAF-DB (Li et al., 2017), CK+ (Lucey et al., 2010), FER-2013 (Goodfellow et al., 2013), RAVDESS (Livingstone and Russo, 2018).